Forward SDE (data to noise):

$$dX_t = f(X_t, t)dt + g(t)dW_t, \quad X_0 \sim p_{data}.$$

In practice, f, g are usually chosen as below:

$$f(x,t) = -\frac{1}{2}\beta(t)x, \quad g(t) = \sqrt{\beta(t)}.$$

The training objective is

$$\begin{aligned}\mathcal{J}(\theta) &= \mathbb{E}_{t\sim U(0,T)}\mathbb{E}_{x\sim p_t}\left[\omega(t)\|s_\theta(x,t) - \nabla_x \log p_t(x)\|_2^2\right] \\ &\equiv \mathbb{E}_t\mathbb{E}_{x_0\sim p_0, x_t\sim p_t(\cdot|x_0)}\left[\omega(t)\|s_\theta(x_t,t) - \nabla_{x_t}\log p_t(x_t, x_0)\|_2^2\right].\end{aligned}$$

To compute this loss function and take gradient w.r.t. θ, we need x_t, t pairs and the conditional score $\nabla_{x_t} \log p_t(x_t|x_0)$. With the choice of f, g above, under linear drift, the $X_t|X_0$ is Gaussian

$$X_t = \alpha(t)X_0 + \sigma(t)\varepsilon, \quad \alpha(t) = \exp(-\frac{1}{2}\int_0^t \beta), \sigma(t) = \sqrt{1-\alpha(t)^2}.$$

Then we could compute the conditional score

$$\nabla_{x_t}\log p_t(x_t|x_0) = -\frac{x_t - \alpha(t)x_0}{\sigma(t)^2}.$$

The training weight at each time step $\omega(t)$ is usually chosen to be σ^2 or β. A popular choice of β (used in generation phase) and corresponding α (used in training phase) is the cosine schedule given in (Nichol & Dhariwal, 2021):

$$\begin{aligned} t = 0, 1, \ldots, T, \qquad & f(t) = \cos^2\left(\frac{\pi}{2}\frac{t/T+s}{1+s}\right), \qquad && s = 0.008 \ll 1, \\ & \bar{\alpha}_t = f(t)/f(0), && t = 0, \ldots, T, \\ & \alpha_t = \frac{\bar{\alpha}_t}{\bar{\alpha}_{t-1}}, \qquad \beta_t = 1 - \alpha_t, && t = 1, \ldots, T. \end{aligned}$$

Since $\nabla_x \log p_t(x)$ is unknown, we use the denoising score matching (DSM) equivalence. Backward SDE:

$$\begin{aligned} dX_t &= \left[-f(X_t,t) + \frac{1+\eta^2}{2}g(t)^2 s_\theta(X_t,t)\right]dt + \eta g(t)dW_t, \quad t: T \to 0. \\ x_{k+1} &= x_k + \left[-f(x_k,t_k) + \frac{1+\eta^2}{2}g(t_k)^2 s_\theta(x_k,t_k)\right]\Delta t + \eta g(t_k)\sqrt{-\Delta t}z_k, \quad z_k \sim \mathcal{N}(0, I_d), \end{aligned}$$

where $\Delta t < 0, T = t_0 > t_1 > \cdots > t_N = 0$. Initialize x_T from a standard Gaussian. If $\eta = 1$, then we use stochastic reverse-time SDE (maximum diversity). If $\eta = 0$, we use probability flow ODE (deterministic, but often fastest if employing external ODE solvers).